plantDescriptionMonitor

Service Description

Service ID: *"plantDescriptionMonitor"*

Abstract
This document describes the `plantDescriptionMonitor` service, version 1.0.0, produced by the PlantDescriptionEngine support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `plantDescriptionMonitor` service, which provides read access to the plant descriptions enriched with the matched registered systems and their monitoring state, and to the alarms raised when the running cloud differs from the description. It is produced by the PlantDescriptionEngine system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

Successor of the Arrowhead 4.6 PlantDescriptionEngine monitor service; liveness now comes from SysMonitor instead of pinging `monitorable` services.

2.2 How This Service Is Meant to Be Used

Operators and dashboards read descriptions and alarms and acknowledge alarms.

2.3 Important Delimitations and Dependencies

Monitoring data are only as recent as SysMonitor's history. With `service.authorization.enabled=true` every call is checked with ConsumerAuthorization.

3 Service Interface

This section describes the interfaces to the `plantDescriptionMonitor` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

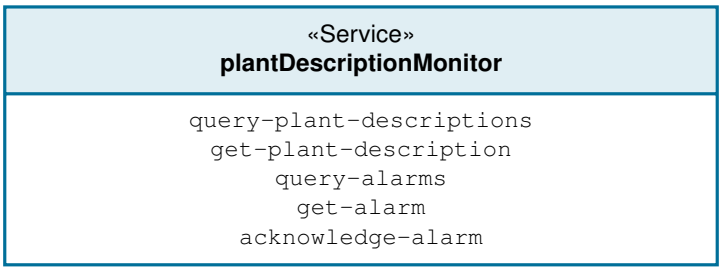


Figure 1: SysML-style block of the `plantDescriptionMonitor` service and its operations.

The following interface operations are available.

3.1 operation query-plant-descriptions () : MonitorPlantDescriptionListResponse

Returns a page of plant descriptions with monitoring data.

3.2 operation get-plant-description (Identifier) : MonitorPlantDescriptionResponse

Returns a plant description with monitoring data.

3.3 operation query-alarms () : PdeAlarmListResponse

Returns a page of alarms, filtered by system name, severity and acknowledgement.

3.4 operation get-alarm (Identifier) : PdeAlarmResponse

Returns an alarm.

3.5 operation acknowledge-alarm (PdeAlarmUpdate) : PdeAlarmResponse

Acknowledges an alarm.

4 Information Model

Here, all data objects that can be part of the `plantDescriptionMonitor` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.10, which are used to represent things like hashes and identifiers.

4.1 struct Connection

A connection from a consumer port to a producer port.

Field	Type	Description
consumer	SystemPort	The consumer port of the connection.
producer	SystemPort	The producer port of the connection.
priority	Number	Priority of the resulting orchestration rules (1 highest).

4.2 struct MonitorPlantDescriptionListResponse

List of plant descriptions with monitoring data.

Field	Type	Description
entries	List<MonitorPlantDescription Response>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.3 struct MonitorPlantDescriptionResponse

A plant description with monitoring data.

Field	Type	Description
id	Number	Identifier of the plant description.
plantDescription	String	Name of the plant description.
active	Boolean	Whether the entry is in force.
include	List<Number>	Identifiers of included plant descriptions.
systems	List<MonitorSystem>	Systems to create or update.
connections	List<Connection>	Connections between consumer and producer ports.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.4 struct MonitorSystem

A plant system with its matched registered system and monitoring data.

Field	Type	Description
systemId	Identifier	Identifier of the system within the plant description.
systemName	Name	Name of the system, PascalCase (e.g. `TemperatureSensor`).
metadata	Metadata	Metadata the registered system must have.
ports	List<Port>	Services the system consumes or produces.
registeredSystemName	String	Name of the registered system matched to this plant system.
monitoring	Map<String, Object>	Latest monitoring information from SysMonitor.
lastObservationTime	DateTime	Time of the latest stored observation of the system.

4.5 struct PdeAlarmListResponse

List of alarms.

Field	Type	Description
entries	List<PdeAlarmResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.6 struct PdeAlarmResponse

An alarm.

Field	Type	Description
id	Number	Identifier of the alarm.
systemId	Identifier	Identifier of the system within the plant description.
systemName	Name	Name of the system the alarm concerns.
acknowledged	Boolean	Whether an operator acknowledged the alarm.
severity	String	Severity of the alarm.
cause	String	Alarm cause: `SYSTEM_NOT_REGISTERED`, `MULTIPLE_MATCHES`, `SYSTEM_NOT_IN_DESCRIPTION`, `NOT_MONITORED` or `NO_RECENT_OBSERVATION`.
description	String	Human-readable alarm text.
raisedAt	DateTime	Time the alarm was raised.
updatedAt	DateTime	UTC time of the last change.
clearedAt	DateTime	Time the alarm was cleared, if cleared.
acknowledgedAt	DateTime	Time of the acknowledgement.

4.7 struct PdeAlarmUpdate

Acknowledgement of an alarm.

Field	Type	Description
acknowledged	Boolean	Set to true to acknowledge the alarm.

4.8 struct Port

A service a plant system consumes or produces.

Field	Type	Description
portName	String	Name of the port within its system.
serviceDefinition	Name	Service definition name.
serviceInterface	String	Required interface template of the port, if any.
metadata	Metadata	Metadata the service instance must have.
consumer	Boolean	True for a consumed service, false for a produced one.

4.9 struct SystemPort

Reference to a port of a plant system.

Field	Type	Description
systemId	Identifier	Identifier of the system within the plant description.
portName	String	Name of the port within its system.

4.10 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Metadata	A set of key-value pairs with arbitrary, possibly nested values.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
Object	Any structured value.
String	A chain of characters.

5 References

[1] J. Delsing, “PlantDescriptionEngine – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-PlantDescriptionEngine-java-spring/documentation/v5.3.0/PlantDescriptionEngine-v5.3.0-SysD.pdf.

[2] ——, “plantDescriptionMonitor HTTP/TLS/JSON – Interface Design Description (PlantDescriptionEngine),” Eclipse Arrowhead, IDD 5.3.0, 2026, plantDescriptionEngine-v5.3.0-plantDescriptionMonitor-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-23	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	